



UTM

UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER 1

2024/2025

SECR1213-02 NETWORK COMMUNICATIONS

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GROUP NAME: Y-LINK

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1.0 Identify the work areas on the floor plan

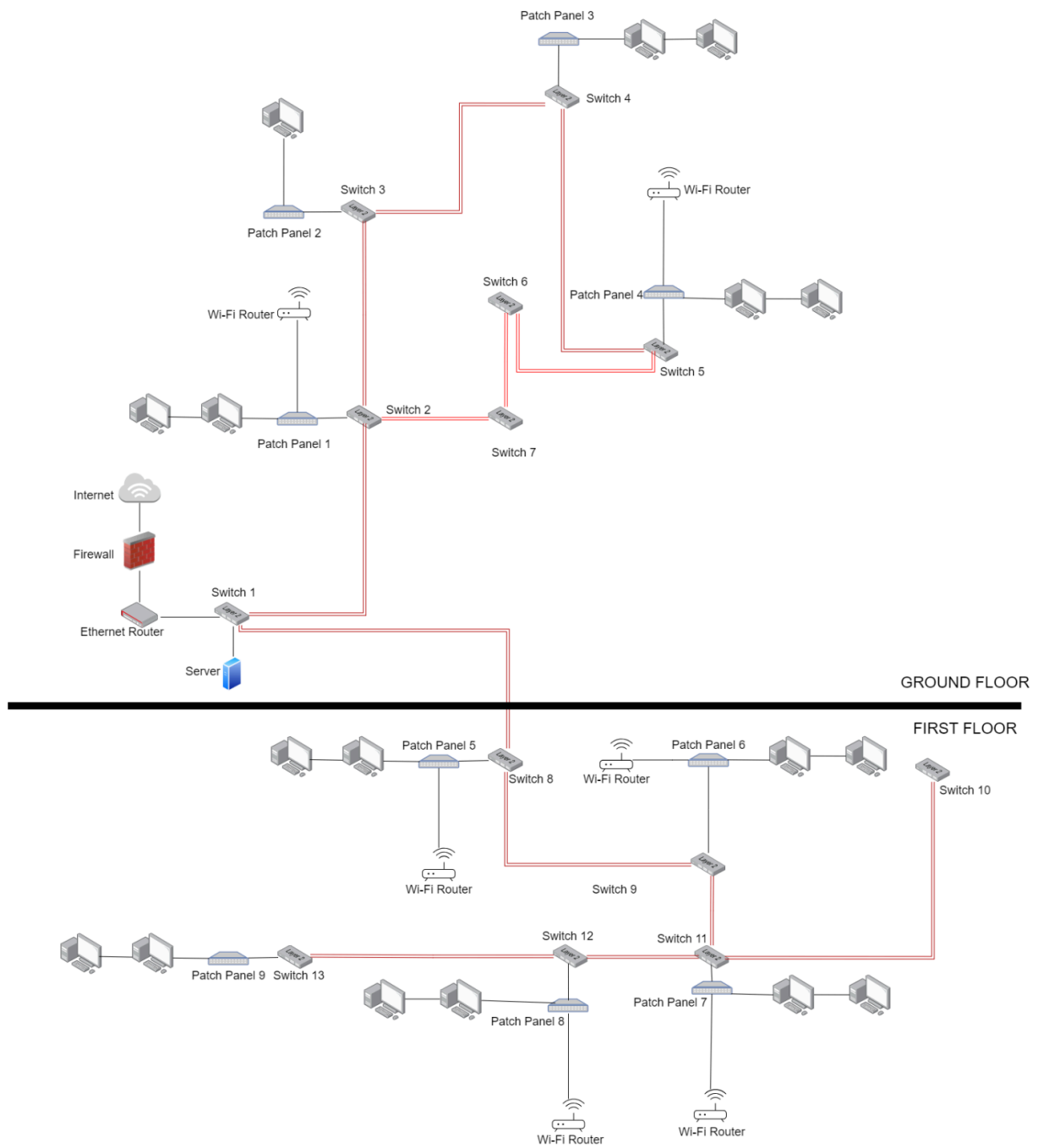
Ground Floor



First Floor

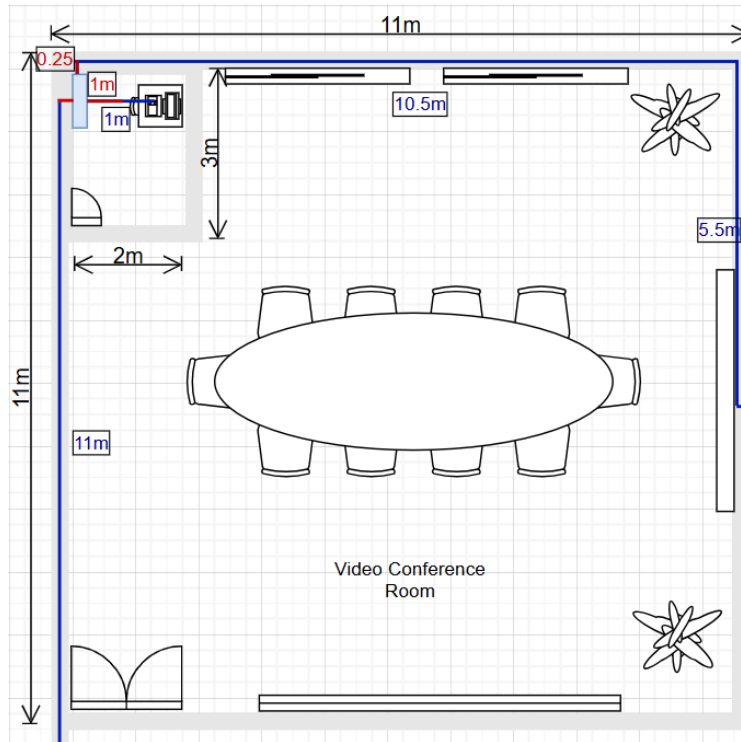


1.1 Overall Network Diagram of the Building



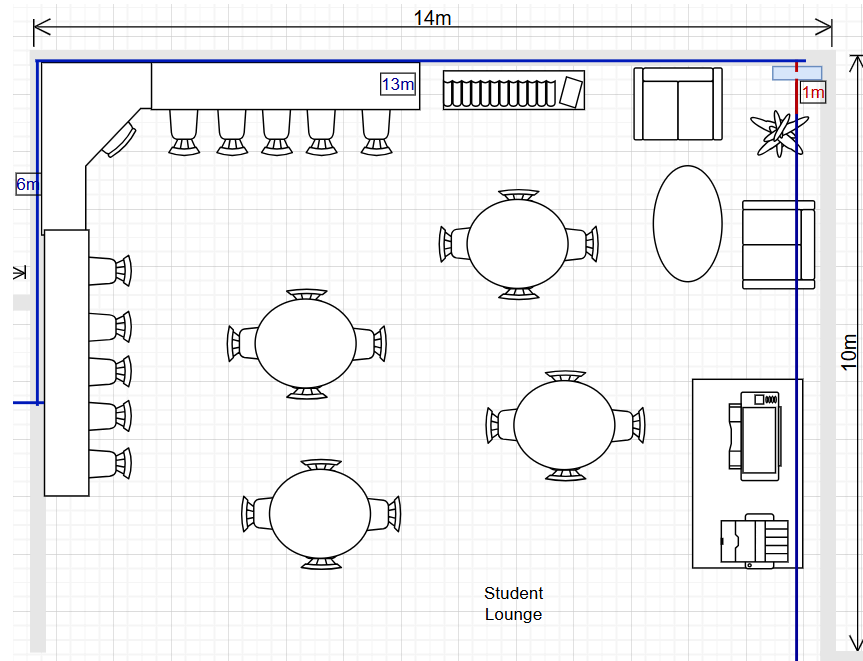
Work Area 1 : Video Conference Room

The video conference room is spacious, square space measuring 11 by 11 meters. It has a large oval table with six chairs in the center and two windows for natural light. The room is equipped with wireless Wi-Fi, allowing easy connectivity with smartphones, laptops and other devices.



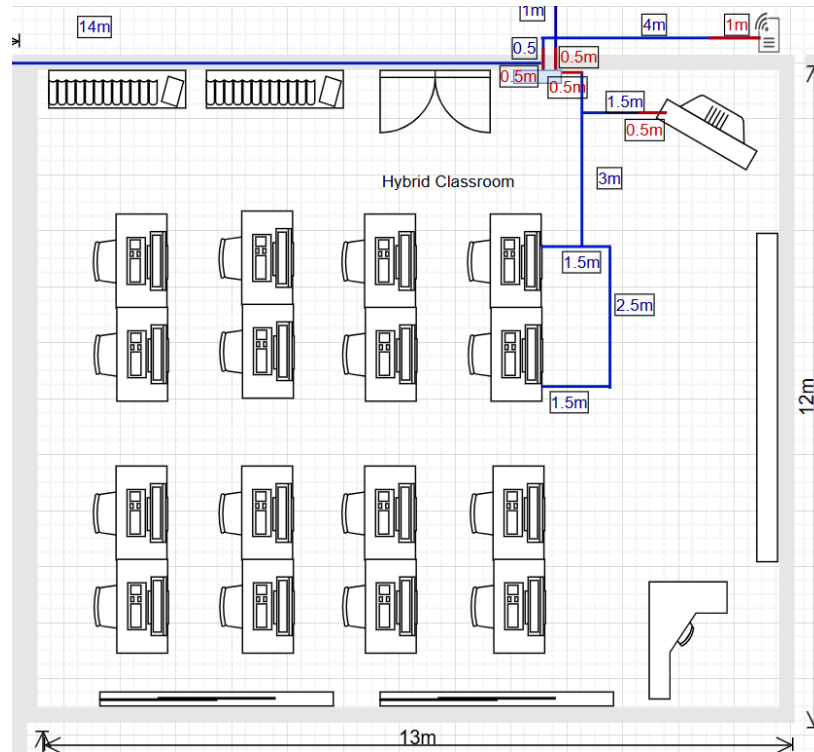
Work Area 2 : Student Lounge

The student lounge is a cozy and versatile space measuring 14 by 10 meters. It has a counter for casual seating or studying, round tables for group work, and a relaxing area. The lounge features wireless Wi-Fi, enabling students to connect their smartphones, laptops, and tablets while doing work or group discussions.



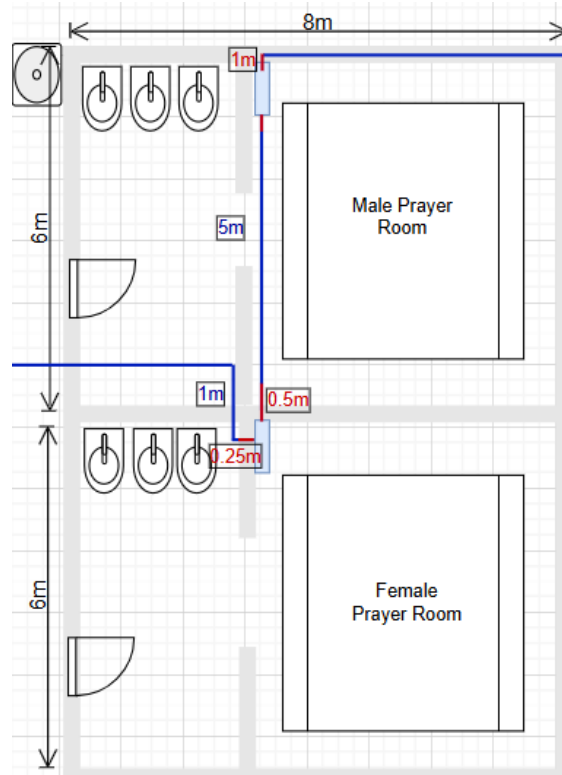
Work Area 3 : Hybrid Classroom

This Hybrid Classroom is a 12m x 13m space fitted out with several workstations in rows. Each of the workstations would include seating and computers, creating an environment suitable for collaborative or individual work. A projector or screen is positioned near the top right corner, indicating support for presentations or virtual learning. The space is likely used for flexible educational activities, blending traditional and digital tools.



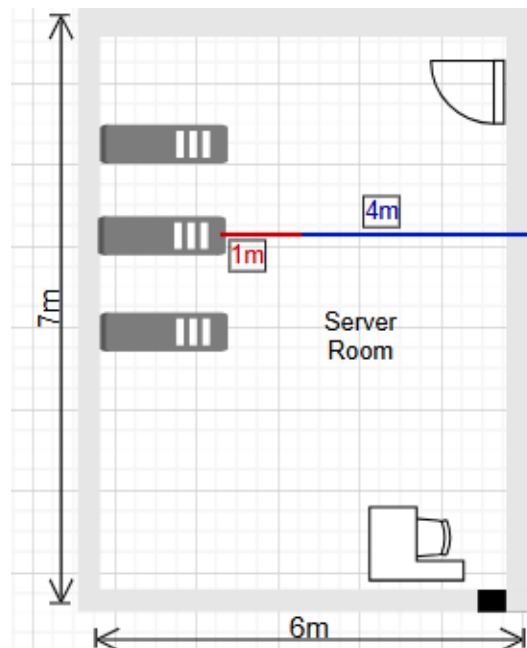
Work Area 4 : Prayer Room

The Prayer Room is divided into two areas, one for male and one for female, approximately 6m x 4m each. Each of these areas has an ablution area with three taps. The Prayer Room are designed to be entered through different doors to ensure privacy.



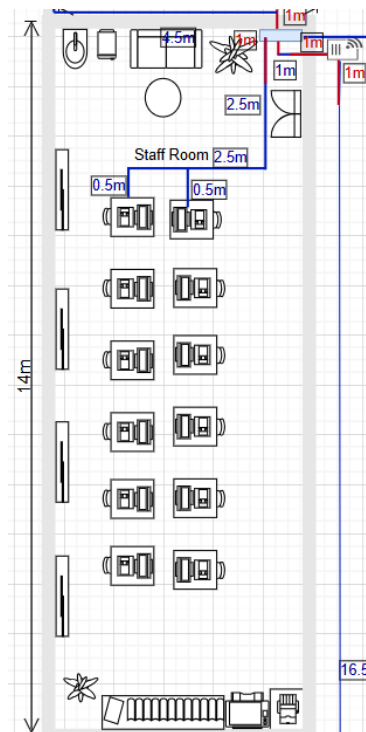
Work Area 5 : Server Room

The server room with the measuring of 6m x 7m has three server racks positioned along one side and a desk in the opposite corner. There is a single door at the top right corner of the room. This design allows ample room for the maintenance and operation of servers.



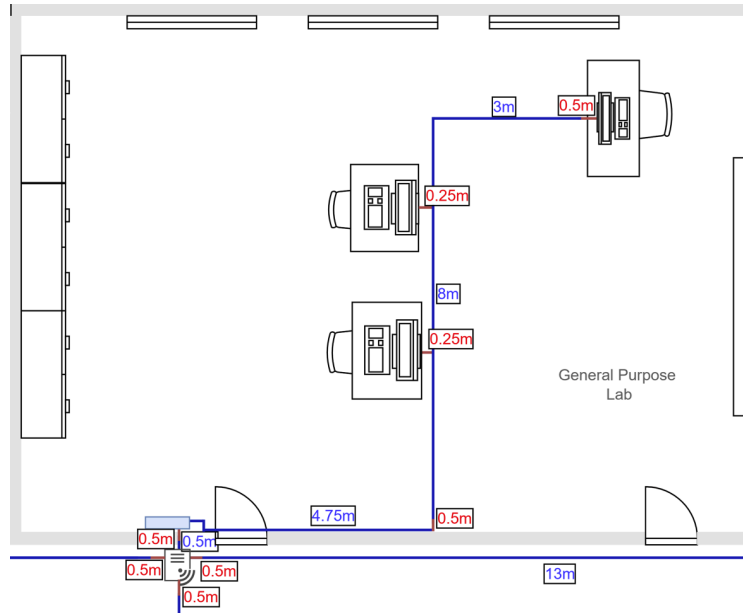
Work Area 6 and 11 : Staff room

Staff Room is a place where the staff and lecturers do their work. There are some workstations provided which are connected to the switch. They can connect to the Internet using their devices like laptops and smartphones because there is a wireless router to provide network connection.



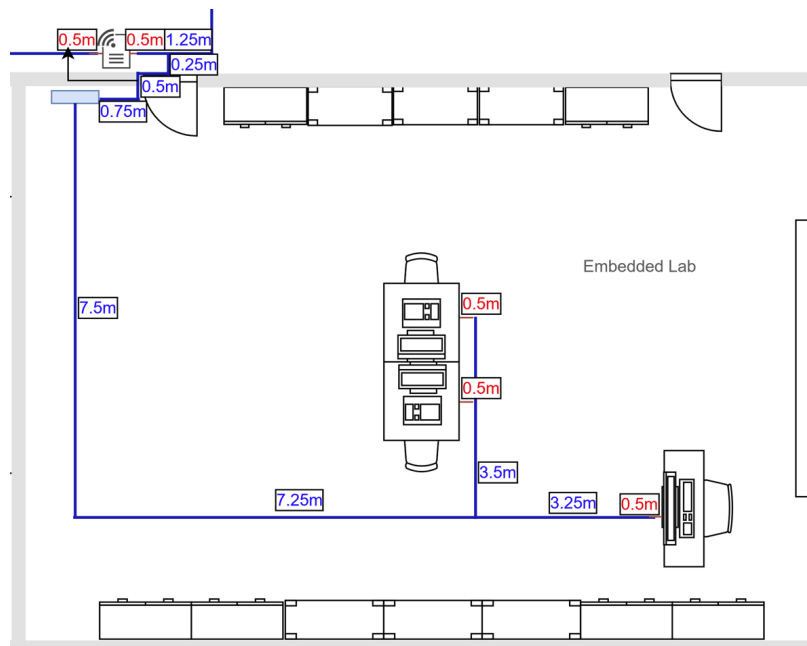
Work Area 7 and 8 : General Purpose Lab

The General Purpose Lab includes multiple rows of desks with computers, specifically 30 workstations arranged in a way that facilitates both individual work and collaborative activities. Each desk is equipped with a chair and is positioned to allow easy movement and access throughout the lab. Additionally, there is a dedicated instructor's area located at the front, featuring a larger desk and equipment for presentation purposes.



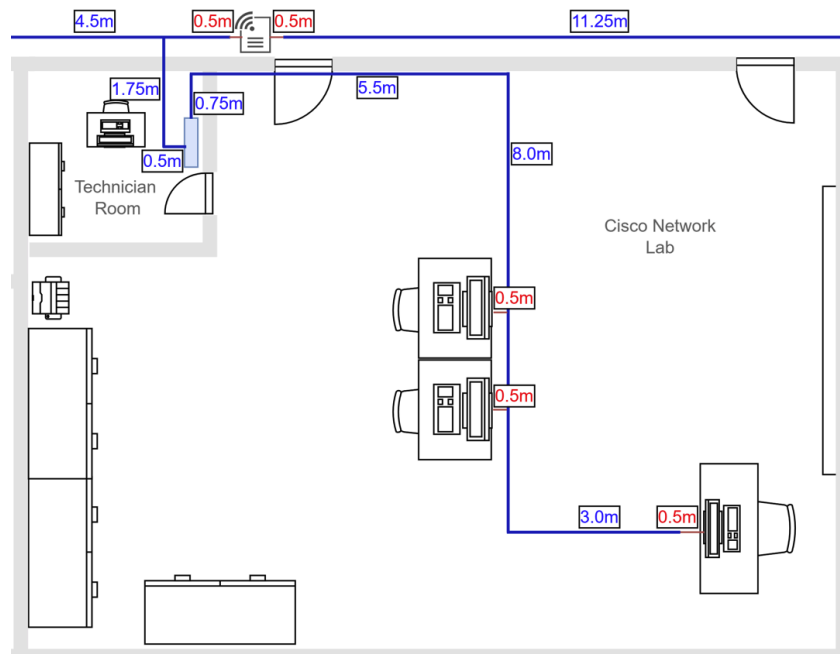
Work Area 9 : Embedded Lab

The Embedded Lab provides ample space for multiple users with the room size of 14x10 meters. Inside, four rows of workstations are arranged neatly, each equipped with desks and chairs. At the side of the room, a dedicated equipment area can be used to house critical networking hardware such as routers, switches, and servers, essential for hands-on practice and simulations. With two doors positioned at opposite ends, the lab ensures smooth entry and exit.



Work Area 10 : Cisco Network Lab

The Cisco Network Lab features multiple rows of workstations arranged around the perimeter and center with the size room of 14x10 meters. Each workstation includes a computer and ample desk space for students to set up and connect peripherals or embedded devices. At the center of the room is a dedicated area with a larger work table and monitor, ideal for demonstrations or group projects. The instructor's desk is positioned at the front of the lab, equipped with additional presentation tools. A separate equipment storage area is located at the bottom left. A technician room is located at the top left, ensuring technical support is available if needed.



2.0 How many connections, patch cords and switch ports have you determined you need?

Switch Port

Work Area	Number of Switch Ports Needed
Video Conference Room	5
Student Lounge	2
Hybrid Classroom	24
Prayer Room	2
Server Room	12
Staff Room (GF)	14
Staff Room (FF)	45
General Purpose Lab	62
Embedded Lab	31
Cisco Network Lab	32
Total	229

Patch Cord

Work Area	Number of Patch Cord Needed
Video Conference Room	5
Student Lounge	2
Hybrid Classroom	24
Prayer Room	2
Server Room	12
Staff Room (GF)	14
Staff Room (FF)	45
General Purpose Lab	62
Embedded Lab	31

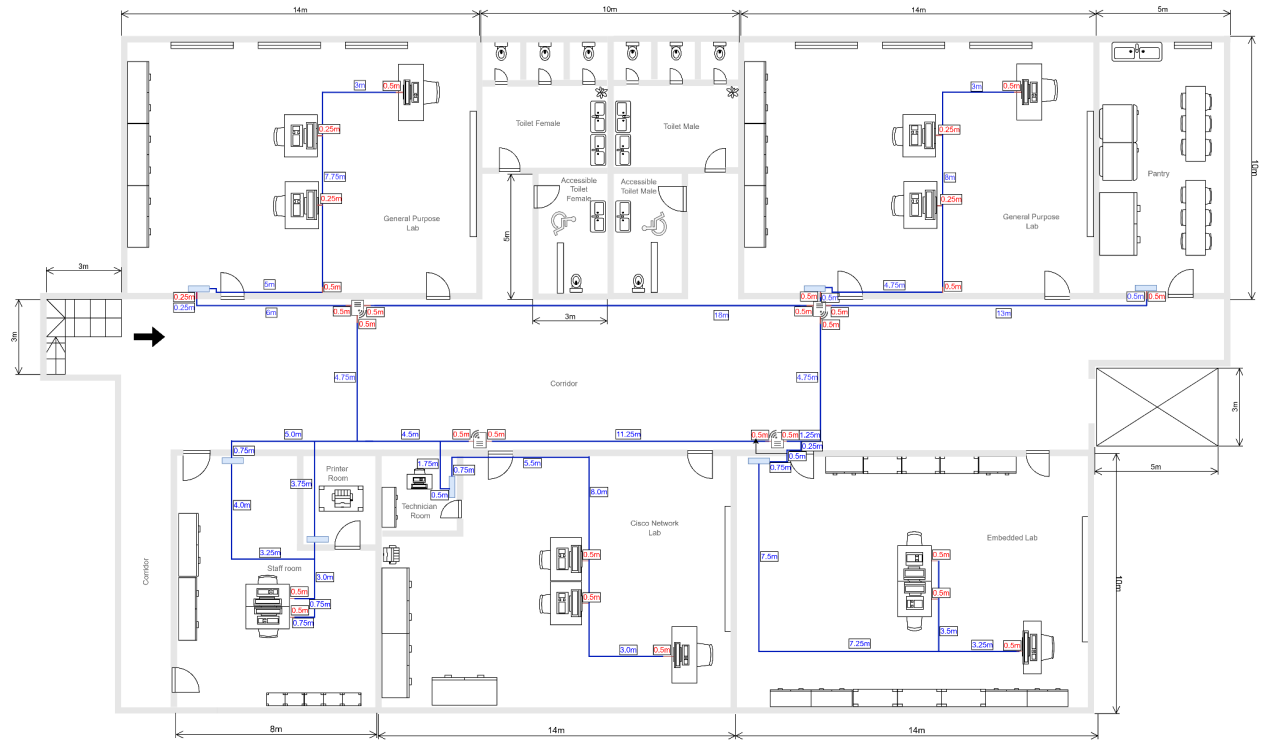
Cisco Network Lab	32
Total	229

Connector

Work Area	Number of Connector Needed
Video Conference Room	10
Student Lounge	4
Hybrid Classroom	48
Prayer Room	4
Server Room	24
Staff Room (GF)	28
Staff Room (FF)	90
General Purpose Lab	124
Embedded Lab	62
Cisco Network Lab	64
Total	458

- First Floor

The first floor will use Cat 6 twisted pair cables to connect between labs, staff room as well as pantry. These cables will also be linked between workstations, switches and the patch panels in labs and rooms.



Cable Type

Cat 6 Cable

For both horizontal and vertical cabling we've opted for Cat 6 twisted-pair cable. Cat 6 cable supports data transfer rates of up to 1 Gbps over 100 meters and 10 Gbps over shorter distances up to 55 meters. Cat 6 cables also support bandwidths of up to 250 MHz which enables fast and reliable data transmission. Thus, Cat 6 cables are efficient to be used as there won't be delays in upload and download of information throughout the building. Apart from that, Cat 6 cables are also Compatible with modems, computers, laptops, networking switches, routers and printers. The suggested floor plan for the building has all these devices, primarily hundreds of computers. Hence, using Cat 6 cable enables all the computers to be connected to the other networking devices such as modems, networking switches and routers. Cat 6 cable is a twisted-pair cable which is designed to minimise electromagnetic interference. Reducing electromagnetic interference ensures that the signal transmitted through cables is intact with minimal distortion which ensures accurate data transmission and reduces errors. The proposed building does not have high-interference environments. So, using Unshielded Twisted-Pair (UTP) cables will be adequate and cost-efficient.

Cable Length

In this project, the cable that we are going to use is CAT6 twisted cables. CAT6 twisted cables are used in the ground floor and first floor according to the floor plan stated above. It will be used to connect all the servers in every work area throughout the whole building.

Work Area	Cable Type	Length(meter)
Ground Floor		
Video Conference Room (Horizontal)	CAT6 twisted cable	28.00
Video Conference Room (Vertical)	CAT6 twisted cable	1.25
Student Lounge (Horizontal)	CAT6 twisted cable	33.5
Student Lounge (Vertical)	CAT6 twisted cable	1.00
Hybrid Classroom (Horizontal)	CAT6 twisted cable	33.5
Hybrid Classroom (Vertical)	CAT6 twisted cable	2.00
Prayer Room (Horizontal)	CAT6 twisted cable	6.00
Prayer Room (Vertical)	CAT6 twisted cable	1.75
Staff Room (Horizontal)	CAT6 twisted cable	12.00
Staff Room (Vertical)	CAT6 twisted cable	2.00
Server Room (Horizontal)	CAT6 twisted cable	12.00
Server Room (Vertical)	CAT6 twisted cable	3.00
Horizontal Cabling	CAT6 twisted cable	68.50
Vertical Cabling	CAT6 twisted cable	3.00
Ground Floor Total Length :		210.50

First Floor		
2 General Purpose Lab (Horizontal)	CAT6 twisted cable	56.75
2 General Purpose Lab (Vertical)	CAT6 twisted cable	19.25
Embedded Lab (Horizontal)	CAT6 twisted cable	15.00
Embedded Lab (Vertical)	CAT6 twisted cable	17.00
Cisco Network Lab (Horizontal)	CAT6 twisted cable	22.75
Cisco Network Lab (Vertical)	CAT6 twisted cable	10.50
Staff Room (Horizontal)	CAT6 twisted cable	15.75
Staff Room (Vertical)	CAT6 twisted cable	16.75
First Floor Total Length :		173.75
Both Floor Total Length :		384.25

4.0 References

<https://drive.google.com/drive/folders/1SzdTX4d9gPns2q2aWZWYs47uE8IGsXLc?usp=sharing>

MINUTES MEETING

Date/Time	11 Dec 2024, 1.25 p.m.		
Location	Online (Google Meet)		
Agenda	1. Discuss the questions on Task 4		
	2. Task distribution for each person		
Meeting Host	Harini		
Attendance			
Name	Time	Reason For Absence	
Dayang Farah Farzana	1.25 p.m.	-	
Farra Nurzahin	1.25 p.m.	-	
Harini	1.25 p.m.	-	
Safiya Nursyahadah	1.25 p.m.	-	
Minutes			
No.	Item Discussed	Details	Person in charge
1.	A short discussion on question given on Task 4	- All members read the question and rubric together.	All members
2.	Identify the work areas	- Safiya identified the work areas of the building. - All the members discuss the area that needs to be labeled.	Safiya
3.	Discuss distribution of connection on every work areas	- Dayang and Safiya responsible for the connections in the ground	All members

		floor work area. - Farra and Harini responsible for the connections in the first floor work area.	
4.	Identify cable type	- We identify to use a CAT6 twisted pair cable type for the building based on the Task 3.	Harini
5.	Discuss on how to measure the cable length	- We agreed to use 1cm = 1m cable length.	All members
6.	Meeting ended	- At 1.50 p.m., Harini ends the meeting after all discussions have been done.	Harini